School of Computing
Queen's University

CISC235

Winter 2026

Assignment 1

Deadline: Jan 24, 2026, 11:59 PM

Membership Problem

Given a set S of n numbers and a target value x, the membership problem is to answer the question "Is x in S?" There are numerous ways to answer this question – some are quite sophisticated, but in this assignment we will consider the two simplest to demonstrate the important of Data Structures as explained in the classroom.

Algorithm A: Store the set S in an array or list. Search the list from beginning to end. If $S[i] == x$ for any value of i, stop searching and return "Yes", otherwise return "No"

Algorithm B: Store the set S in an array or indexed list (such as the Python list structure). Sort the set. Use binary search to determine if x is in S. Return "Yes" or "No" as appropriate.

If we are doing a single search, Algorithm A is clearly preferable. However, if we are doing many searches of the same set, Algorithm B is preferable since we only need to sort the set once. Your task is to explore the relationship between the size of the set and the number of searches required to make Algorithm B preferable to Algorithm A.

Implement both Algorithm A and Algorithm B. When implementing Algorithm B, you must write your own sort function and your own binary search function. You may use any sort algorithm that has complexity in O(n*log n).

Add appropriate clock functions to your implementations so that you can measure the actual running time of the two algorithms.

For n = 1000, 2000, 5000, and 10000, conduct the following experiment:

Use a random number generator to create the set S containing n integers (duplicate numbers in the set are ok)
For increasing values of k
choose k target values, half of which are in ***S*** and half are not in ***S***
Using Algorithm A, perform the k searches of ***S***
Using Algorithm B, perform the k searches of ***S***

Let k* be the approximate smallest value of k for which Algorithm B becomes preferable to Algorithm A – note that k* may be greater than n.

Note that the execution time of a function is subject to slight variation due to background activity – you should repeat each experiments at least three times and average the results.

A ***n*** increases, does the ratio of k*/n increase, decrease or remain relatively constant?

Report your findings in a text file. Submit your program listing and your text file.